# Product Requirement Document (PRD) — CliniFlow AI

Team Name : Dependency Free

Participant Name : Tang Shao Xian

# Table of Contents


**1. Project Overview** **3**
**2. Background & Business Objective** **5**
2.1 Background 5
2.2 Business Objectives 5
**3. Product Purpose** **7**
**4. System Functionalities** **8**
4.1 Pre-Visit Phase: AI Symptom Intake Agent 8
4.2 Visiting Phase: AI Clinical Documentation 8
4.3 Post-Visit Phase: Patient Communication & Follow-Up 9
4.4 Core Platform Features 10
**5. User Stories & Use Cases** **11**
5.1 Patient User Stories 11
5.2 Doctor User Stories 11
5.3 Clinic Staff /Receptionist User Stories 12
5.4 Clinic Administrators / Owners 13
5.4 Use Cases 13
**6. Features Included (Scope Definition)** **14**
6.1 Pre-visit 14
6.2 Visiting 14
6.3 Post-Visit 14
**7. Features Not Included (Scope Control)** **15**
**8. Assumptions & Constraints** **16**
8.1 Assumptions 16
8.2 Constraints 17
**9. Risks & Questions Throughout Development** **18**
9.1 Risks 18
9.2 Open Questions 19

# 1. Project Overview

CliniFlow AI is an AI-powered clinical workflow platform designed to streamline and automate the end-to-end patient journey for small and medium-sized (SME) clinics in Malaysia. The system integrates intelligent agents across pre-visit, visiting, and post-visit phases to reduce manual workload, improve documentation quality, and enhance patient care.

The platform is built to support key stakeholders within a clinic ecosystem. Patients interact with an AI-driven symptom intake agent before their visit, enabling structured data collection and better preparation for consultations. During the consultation, clinic doctors utilize the system to automatically generate structured medical documentation from text or voice inputs, significantly reducing time spent on administrative tasks. Clinic staff and receptionists benefit from streamlined intake and workflow processes, while clinic administrators or owners gain better oversight of operations through organized records and system-level visibility.

By transforming unstructured clinical interactions into structured, actionable outputs, CliniFlow AI minimizes inefficiencies caused by fragmented systems and repetitive workflows. The platform focuses on delivering a seamless and intelligent workflow that connects all stakeholders, ensuring that information flows consistently from initial patient intake to post-visit care.

Ultimately, CliniFlow AI aims to empower SME clinics with a scalable, AI-driven solution that enhances operational efficiency, improves documentation accuracy, and elevates the overall patient experience.

**Key Stakeholders:**

| Role | Description |
|---|---|
| Clinic Doctors | Primary users who conduct consultations and review AI-generated documentation |
| Clinic Staff / Receptionists | Manage patient scheduling, intake, and administrative tasks |

| | |
|---|---|
| Patients | Interact with symptom intake agent and review post-visit records |
| Clinic Administrators / Owners | Oversee operations, manage user accounts, and view analytics |

# 2. Background & Business Objective

## 2.1 Background

SME clinics in Malaysia face significant operational challenges due to high patient volumes, limited resources, and reliance on manual processes.

- **Heavy administrative burden:** Doctors spend a large portion of their time on documentation rather than patient care. Studies show clinicians can spend **30–50% of their time on administrative tasks**, which reduces consultation quality and contributes to burnout.
- **Fragmented systems:** Many clinics still rely on a mix of paper records and basic digital tools, leading to inconsistent, duplicated, or incomplete patient data. Unlike large hospitals, SME clinics often lack integrated electronic medical record systems.
- **Repetitive data collection:** Patients are frequently asked to repeat the same information (symptoms, history, allergies) at different stages, causing inefficiency and frustration.
- **Poor patient communication:** Due to time constraints and varying health literacy levels in Malaysia, patients often leave consultations without fully understanding medication instructions or follow-up care, leading to low adherence and poorer outcomes.

Overall, these challenges highlight the need for a solution that reduces administrative workload, improves data consistency, and enhances communication between doctors and patients.

## 2.2 Business Objectives

| | Objective | Success Metric |
|---|---|---|
| 1 | Reduce documentation time for doctors | ≥ 50% reduction in time spent on clinical note-taking per consultation |

| | | |
|---|---|---|
| 2 | Improve data accuracy and completeness | ≥ 90% of patient records contain all required structured fields after each visit |
| 3 | Enhance patient understanding | ≥ 80% of patients report clear understanding of post-visit instructions (via survey) |
| 4 | Streamline clinic workflows | ≥ 40% reduction in average patient check-in to consultation-ready time |
| 5 | Centralize patient records | 100% of patient records accessible from a single platform within 6 months of deployment |

# 3. Product Purpose

CliniFlow AI serves as an **end-to-end intelligent workflow agent** that bridges the gap between clinical care and administrative efficiency. Its purpose is to:

a. **Collects and structures patient data before consultation** —CliniFlow AI enables structured data collection during the pre-visit phase through an AI-driven symptom intake agent. Instead of relying on unstructured or incomplete patient descriptions, the system guides patients to provide relevant information and organizes it into a clear, standardized format.

b. **Automate Clinical Documentation During Consultation** — The platform assists doctors by automatically generating structured medical documentation from consultation inputs, including text and voice recordings.

c. **Ensure Consistent and Structured Medical Records** —CliniFlow AI centralizes patient information into a unified system, ensuring that all records are stored in a structured and consistent format.

d. **Improve Patient Understanding and Post-Visit Care** — After the consultation, the system generates clear and simplified instructions for patients, including medication details, dosage, and care guidelines.

**Value Proposition:**

- **For Doctors:** Less time documenting, more time caring. AI handles the paperwork.
- **For Patients:** Better understanding of their health, clearer instructions, and a more seamless clinic experience.
- **For Clinic Owners:** Improved operational efficiency, reduced administrative costs, and higher patient satisfaction.

# 4. System Functionalities

The system is divided into three phases:

## 4.1 Pre-Visit Phase: AI Symptom Intake Agent

| Feature | Description |
|---|---|
| Conversational Symptom Collection | AI chatbot engages patients in a guided conversation to collect symptoms, medical history, allergies, and current medications before the consultation. Text and voice input are both supported. |
| Multi-language Support | Supports English and Bahasa Malaysia for patient interaction |
| Structured Pre-Visit Report | Automatically generates a structured symptom summary report for the doctor to review before the consultation |
| Patient Profile Auto-Population | Pre-fills known patient information from previous visits to avoid repetitive data entry |

## 4.2 Visiting Phase: AI Clinical Documentation

| Feature | Description |
|---|---|
| Multi-Modal Input | Supports both text-based note entry and voice recording during consultations, allowing flexibility based on doctor preference and workflow. |
| Speech-to-Text Transcription | Converts doctor-patient conversations into text, enabling automatic capture of consultation details for further processing. |

| | |
|---|---|
| Real-Time / Post-Consultation Processing | Allows documentation to be generated either in real-time during consultation or after the session using recorded inputs. |
| AI-Structured Clinical Notes | Automatically organizes transcribed or typed inputs into structured formats such as SOAP (Subjective, Objective, Assessment, Plan) to standardize documentation. |
| Medical Entity Extraction | Identifies and extracts key clinical information such as symptoms, duration, medications, and observations from unstructured inputs. |
| Missing Information Detection | Detects incomplete or missing critical information (e.g., allergies, duration of symptoms) and prompts the user to provide additional details. |
| Centralized Patient Records | Stores all consultation data in a unified patient record system accessible to authorized clinic staff and patients. |
| Doctor Review & Editing | Allows doctors to review, edit, and validate AI-generated documentation before finalizing records. |
| Patient Record Access | Enables patients to view their consultation records through the platform after the visit. |
| Data Structuring & Standardization | Ensures all captured data follows a consistent structure, improving readability, retrieval, and future reference. |

## 4.3 Post-Visit Phase: Patient Communication & Follow-Up

| Feature | Description |
|---|---|
| Patient-Friendly Summary | AI generates a clear, non-technical summary of the consultation including diagnosis, prescribed medications, dosage instructions, and follow-up schedule |

| | |
|---|---|
| Medication & Dosage Instructions | Structured medication list with dosage, frequency, duration, and special instructions |
| Patient Portal Access | Patients can log in to review their visit history, summaries, and instructions at any time |

## 4.4 Core Platform Features

| Feature | Description |
|---|---|
| User Authentication & Role Management | Secure login with role-based access control (Doctor, Staff, Patient, Admin) |
| Patient Records Management | CRUD operations for patient demographics, medical history, and visit records |
| Dashboard & Analytics | Overview of clinic operations including patient volume, consultation times, and documentation metrics |

# 5. User Stories & Use Cases

## 5.1 Patient User Stories

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-P01 | As a patient, I want to describe my symptoms before visiting so that the doctor understands my condition in advance | - Patient can input symptoms in text form<br>- System accepts unstructured input<br>- Input is successfully submitted and processed |
| US-P02 | As a patient, I want the system to ask follow-up questions so that my symptoms are captured accurately | - System detects missing information (e.g., duration, severity)<br>- Follow-up questions are generated dynamically<br>- Patient can respond and update information |
| US-P03 | As a patient, I want to view my medical record after the visit so that I understand what was discussed | - Patient can access their record<br>- Record displays structured consultation details<br>- Data is readable and complete |
| US-P04 | As a patient, I want to receive clear medication instructions so that I can follow treatment correctly | - System generates medication details<br>- Dosage and frequency are included<br>- Instructions are presented in simple language |
| US-P05 | As a patient, I want to understand my care instructions so that I can follow up properly | - Post-visit summary is generated<br>- Instructions are clear and concise<br>- Key actions (e.g., follow-up) are included |

## 5.2 Doctor User Stories

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-D01 | As a doctor, I want to receive a structured pre-visit report so that I can prepare before consultation | - System generates a structured report (symptoms, duration, history)<br>- Report is accessible before consultation<br>- Report format is clear and readable |

| | | |
|---|---|---|
| US-D02 | As a doctor, I want to record consultation details using text or voice so that I can capture information efficiently | - System supports text input<br>- System accepts voice recording (or simulated input)<br>- Input is successfully stored |
| US-D03 | As a doctor, I want the system to convert conversations into structured notes so that I don't need to write manually | - Speech/text input is processed by the system<br>- Structured notes (e.g., SOAP format) are generated<br>- Output is displayed clearly |
| US-D04 | As a doctor, I want to review and edit AI-generated notes so that I can ensure accuracy before saving | - Generated notes are editable<br>- Doctor can modify content<br>- Final version can be saved |
| US-D05 | As a doctor, I want the system to detect missing information so that important details are not overlooked | - System identifies missing fields (e.g., allergies)<br>- System prompts user for additional input<br>- Updated data is reflected in final report |
| US-D06 | As a doctor, I want patients to have access to clear instructions so that treatment adherence improves | - Instructions are automatically generated after consultation<br>- Patients can access them via the platform |

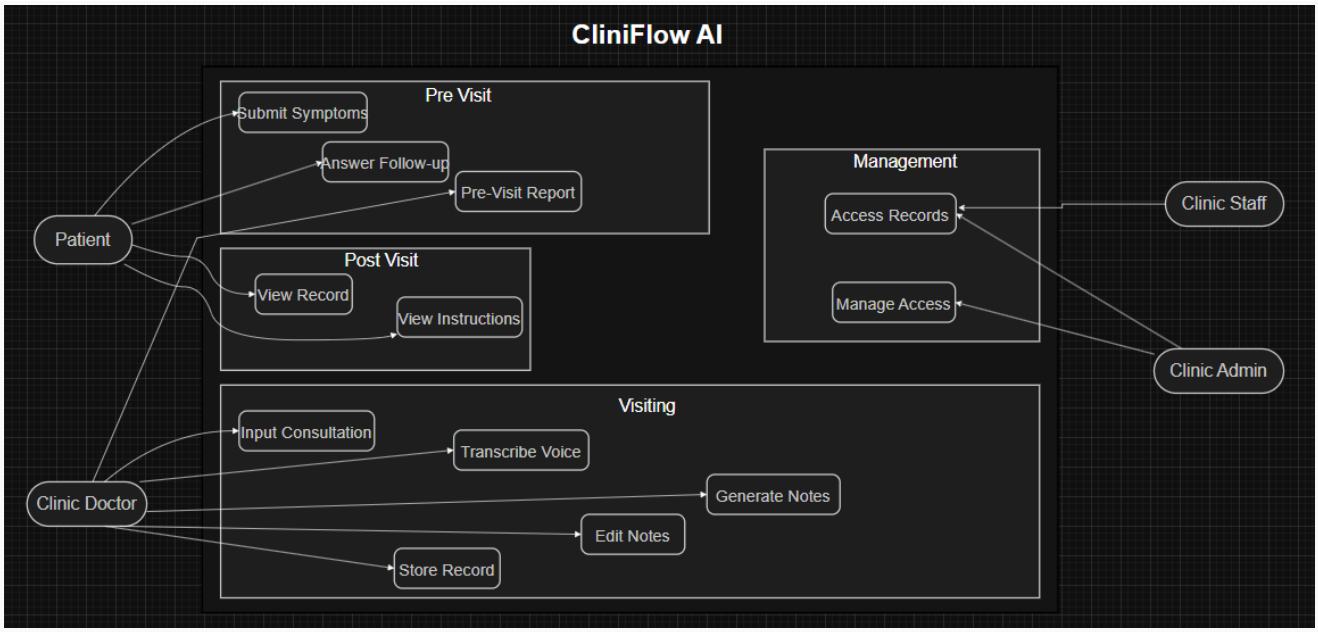
## 5.3 Clinic Staff /Receptionist User Stories

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-R01 | As clinic staff, I want patient information to be collected before the visit so that intake is faster and more efficient | - Pre-visit data is available before appointment<br>- Data is structured and complete<br>- Staff can access patient information |
| US-R02 | As clinic staff, I want access to patient records so that I can assist in administrative tasks | - Staff can retrieve patient records<br>- Data is displayed in structured format<br>- Access is limited to authorized users |

## 5.4 Clinic Administrators / Owners

| ID | User Story | Acceptance Criteria |
|---|---|---|
| US-O01 | As an administrator, I want a centralized system for patient records so that clinic operations are more organized | - All patient data is stored in one system<br>- Records are consistent and structured<br>- Data is retrievable when needed |
| US-O02 | As an administrator, I want controlled access to the system so that patient data is secure | - Only authorized users can access records<br>- Basic access control is implemented<br>- Unauthorized access is restricted |

## 5.4 Use Cases



# 6. Features Included (Scope Definition)

## 6.1 Pre-visit

| | Features | Priority |
|---|---|---|
| 1 | Symptom Intake Agent | Must have |
| 2 | Pre-visit report generation | Must have |
| 3 | Voice input in symptom collection | Should have |
| 4 | Appointment booking | Should have |

## 6.2 Visiting

| | Features | Priority |
|---|---|---|
| 1 | Multi-input support (text + voice) | Must have |
| 2 | Documentation Agent | Must have |
| 3 | Structured report generation | Must have |
| 4 | Patient record storage | Must have |
| 5 | AI-suggested diagnosis codes and medication auto-complete | Should have |

## 6.3 Post-Visit

| | Features | Priority |
|---|---|---|
| 1 | Patient record viewing | Must have |
| 2 | Instruction generation (medication, dosage, etc.) | Must have |
| 3 | Structured report generation | Must have |
| 4 | Patient record storage | Must have |
| 5 | Follow-up appointment reminders (in-app notifications) | Should have |

| | | |
|---|---|---|
| 6 | Medication schedule reminders | Should have |
| 7 | Operational analytics dashboard for clinic admins | Should have |

# 7. Features Not Included (Scope Control)

The following features are explicitly **out of scope** for this project to maintain focus and feasibility:

| | **Excluded Feature** | **Reason** |
|---|---|---|
| 1 | E-prescribing with direct pharmacy integration | Requires regulatory compliance with Malaysian Pharmacy Board; deferred to future phase |
| 2 | Telemedicine / video consultation | Adds significant complexity in infrastructure and compliance; not part of core workflow automation |
| 3 | Insurance claims processing | Requires integration with Malaysian insurance providers; separate initiative |
| 4 | Medical imaging analysis (X-ray, MRI) | Specialized AI domain outside the scope of workflow automation |
| 5 | Full Electronic Health Record (EHR) system | CliniFlow AI focuses on workflow automation, not replacing full EHR systems; integration with existing EHRs may be considered later |
| 6 | Patient appointment booking / scheduling | Assumed to be handled by existing clinic scheduling tools; CliniFlow AI starts from post-booking |

| | | |
|---|---|---|
| 7 | Billing and payment processing | Separate domain; clinics typically have existing billing systems |
| 8 | Mobile native app (iOS/Android) | Web-responsive design is prioritized; native apps may be considered in future phases |

# 8. Assumptions & Constraints

## 8.1 Assumptions

| | **Assumption** |
|---|---|
| 1 | Target clinics have reliable internet connectivity for web-based access. |
| 2 | Doctors and staff have basic digital literacy and can interact with a web application. |
| 3 | Patients have access to a smartphone or computer to interact with the symptom intake chatbot and patient portal. |
| 4 | Clinics are willing to adopt a new digital workflow tool alongside or replacing their current manual processes. |
| 5 | AI-generated clinical notes will always require doctor review and confirmation before being finalized — the AI assists but does not replace clinical judgment. |
| 6 | English is sufficient for MVP; Bahasa Malaysia support will be added in Phase 2. |
| 7 | Voice transcription accuracy of ≥ 90% is achievable for Malaysian-accented English using available speech-to-text APIs. |

## 8.2 Constraints

| | Constraint | Impact |
|---|---|---|
| 1 | **Data Privacy Compliance:** Must comply with Malaysia's Personal Data Protection Act 2010 (PDPA) for handling patient health information. | Affects data storage, access controls, and consent mechanisms. |
| 2 | **AI Accuracy Limitations:** AI-generated notes and summaries must be treated as drafts requiring human verification; the system must not imply autonomous clinical decision-making. | Requires clear UI indicators that AI output is a suggestion, not a diagnosis. |
| 3 | **Budget Constraints:** SME clinics have limited IT budgets; solutions must be cost-effective with minimal infrastructure requirements. | Cloud-hosted SaaS model preferred; avoid expensive on-premise deployments. |
| 4 | **Timeline:** MVP must be deliverable within 3 months to validate core value proposition with pilot clinics. | Limits Phase 1 scope to essential features only. |
| 5 | **Language/Accent Diversity:**Malaysia has diverse linguistic patterns; voice transcription must handle code-switching (English/Malay mix). | May limit voice feature accuracy in Phase 2; requires iterative improvement. |

# 9. Risks & Questions Throughout Development

## 9.1 Risks

| | Risk | Likelihood | Impact | Mitigation Strategy |
|---|---|---|---|---|
| R1 | AI-generated clinical notes contain inaccuracies that go undetected by the reviewing doctor. | Medium | High | Mandatory doctor review step with explicit confirmation; highlight AI-generated sections with confidence indicators; audit trail for all changes. |
| R2 | Low adoption by doctors who are accustomed to manual workflows or resistant to AI tools. | Medium | High | Conduct user research and co-design sessions with pilot clinic doctors; provide comprehensive onboarding and training; iterate based on feedback. |
| R3 | Voice transcription accuracy is insufficient for Malaysian-accented English and code-switching. | Medium | Medium | Start with text-based input in MVP; evaluate multiple speech-to-text providers; collect voice samples for fine-tuning in Phase 2. |
| R4 | Patient data breach or non-compliance with PDPA. | Low | Critical | Implement encryption at rest and in transit; role-based access control; regular security audits; obtain legal review of data handling practices. |

| | | | | |
|---|---|---|---|---|
| R5 | Scope creep due to clinic-specific customization requests during pilot. | High | Medium | Clearly define MVP scope (Section 6); use feature request backlog; evaluate requests against product roadmap before committing. |
| R6 | Dependency on third-party AI/ML APIs (e.g., LLM, speech-to-text) introduces latency, cost, or availability risks. | Medium | Medium | Abstract AI services behind an internal API layer; implement fallback mechanisms; monitor API costs and performance. |

## 9.2 Open Questions

| | **Question** | **Owner** | **Status** |
|---|---|---|---|
| Q1 | What specific PDPA compliance requirements apply to AI-processed health data in Malaysia? | Legal / Compliance | Open |
| Q2 | Which LLM provider offers the best balance of accuracy, cost, and latency for clinical note generation? | Engineering | Open |
| Q3 | What is the preferred SOAP note format used by Malaysian SME clinics? Are there local variations? | Product / Medical Advisor | Open |
| Q4 | How should the system handle cases where AI confidence in generated notes is low? | Product / UX | Open |
| Q5 | What is the acceptable latency for real-time voice transcription during consultations? | Engineering / UX | Open |

| | | | |
|---|---|---|---|
| Q6 | Will pilot clinics provide access to anonymized historical patient data for AI model training/testing? | Business Development | Open |
| Q7 | What integration points (if any) exist with clinics' current systems (e.g., appointment scheduling, billing)? | Product / Engineering | Open |